Recruiting Agency Database

Business and data model description · Author: Iryna Kapusuz Kutyk

1 BUSINESS DESCRIPTION

1.1 BUSINESS BACKGROUND

Recruiting agencies play an important role in connecting employers with suitable candidates. It works for both sides of HR market. It means they operate a large volume of information such as job offers, applications, interviews, and hiring data daily. All this information needs to be organized and saved in a proper way to avoid operational errors.

1.2 PROBLEMS. CURRENT SITUATION RECRUITING

Before implementing the system agency faced such issues as record dublicates, loss of relevant data, difficulties in linking candidates with employer requirements, possibility of tracking recruitment performance etc. Also with business growing due to operational reasons it became mandatory to organize and standardize data for both sides of customers and agency inner departments.

1.3 THE BENEFITS OF IMPLEMENTING A DATABASE. PROJECT VISION

The database system should let easy and quick access to required information (candidate, employer, job data), make a visible link between candidates and employers, automize tracking of the recruitment process. In addition agency database was designed to store, manage, and analyze necessary data with a single, structured system.

2 MODEL DESCRIPTION

2.1 LOGICAL SCHEME

The logical model follows the **3rd Normal Form (3NF)** to minimize redundancy and maintain data integrity.

Main relationships:

- One **Employer** can have many **Job_Offers**.
- One **Candidate** can apply for many **Jobs** (via the **Application** table).
- One **Application** can have several **Interviews**, and may lead to one **Offer**.
- Each **Candidate** and **Job** can have multiple **Skills** (M:N implemented via Candidate_Skill and Job_Skill).

- Each **Recruiter** manages multiple **Job_Offers** and has performance data stored in Recruiter_Performance.
- Each **Job_Offer** is tied to a specific **Job_Location**.

Relationship	Type	Description
Employer → Job_Offer	1:M	One employer can post many job offers. Each job offer belongs to one employer.
Recruiter → Job_Offer	1:M	One recruiter manages many job offers. Each offer can be assigned to one recruiter.
Job_Location → Job_Offer	1:M	One location can host many job offers.
Job_Offer → Application	1:M	Each job can have many applications.
Candidate → Application	1:M	Each candidate can submit many applications.
Application → Application_Stage	1:M	Each application can pass through several stages (e.g., screening, interview).
Application → Interview	1:M	Each application can have multiple interviews.
Application → Offer	1:1	Each application leads to at most one final job offer.
Recruiter → Recruiter_Performance	1:M	Each recruiter has multiple monthly performance records.
Candidate ↔ Skill (via Candidate_Skill)	M:N	A candidate can have multiple skills; each skill can belong to multiple candidates.
Job_Offer ↔ Skill (via Job_Skill)	M:N	A job can require multiple skills; each skill can be required by multiple jobs.

2.2 OBJECTS

1 CANDIDATE

Field Name	Description	Data Type
Candidate_ID	Unique identifier (PK)	INT
Candidate_First_Name	Candidate’s first name	VARCHAR(100)
Candidate_Last_Name	Candidate’s last name	VARCHAR(100)
Candidate_Email	Contact email (UQ)	VARCHAR(150)

Field Name	Description	Data Type
Candidate_Phone	Phone number	VARCHAR(30)
Candidate_Degree	Education degree	VARCHAR(100)
Candidate_Experience_Years	Work experience in years	DECIMAL(3,1)
Candidate_Is_Active	Candidate status	BOOLEAN

2 EMPLOYER

Field Name	Description	Data Type
Employer_ID	Unique employer ID (PK)	INT
Employer_Name	Company name (UQ)	VARCHAR(150)
Employer_Industry	Field of business	VARCHAR(100)
Employer_Contact_Person	Contact person name	VARCHAR(100)
Employer_Contact_Email	Email	VARCHAR(150)
Employer_Is_Active	Employer status	BOOLEAN

3 RECRUITER

Field Name	Description	Data Type
Recruiter_ID	Unique recruiter ID (PK)	INT
Recruiter_First_Name	Recruiter’s first name	VARCHAR(100)
Recruiter_Last_Name	Recruiter’s last name	VARCHAR(100)
Recruiter_Email	Unique email	VARCHAR(150)
Recruiter_Phone	Phone number	VARCHAR(30)
Recruiter_Hired_On	Hire date	DATE
Recruiter_Is_Active	Employment status	BOOLEAN

4 SKILL

Field Name	Description	Data Type
Skill_ID	Unique skill ID (PK)	INT
Skill_Name	Name of skill (UQ)	VARCHAR(100)
Skill_Category	Category (technical/soft)	VARCHAR(50)
Skill_Is_Active	Active flag	BOOLEAN

5 CANDIDATE_SKILL

Field Name	Description	Data Type
Candidate_ID	Candidate reference (FK)	INT
Skill_ID	Skill reference (FK)	INT
Proficiency_Level	Skill level (1–5)	SMALLINT
Years_With_Skill	Experience duration	DECIMAL(3,1)

6 JOB_SKILL

Field Name	Description	Data Type
Job_ID	Job reference (FK)	INT
Skill_ID	Skill reference (FK)	INT
Requirement_Type	Required/Preferred	VARCHAR(30)
Composite PK	(Job_ID, Skill_ID)	—

7 JOB_LOCATION

Field Name	Description	Data Type
Location_ID	Location ID (PK)	INT
City	City name	VARCHAR(100)
Country	Country name	VARCHAR(100)
Office_Address	Street/office address	VARCHAR(200)
Postal_Code	ZIP code	VARCHAR(20)
Location_Is_Active	Active flag	BOOLEAN

8 JOB_OFFER

Field Name	Description	Data Type
Job_ID	Job ID (PK)	INT
Employer_ID	Employer reference (FK)	INT
Recruiter_ID	Recruiter reference (FK)	INT
Location_ID	Location reference (FK)	INT
Job_Title	Title of position	VARCHAR(150)
Employment_Type	Full-time / Part-time	VARCHAR(30)
Salary	Offered salary	DECIMAL(10,2)
Status	Job status	VARCHAR(30)

9 APPLICATION

Field Name	Description	Data Type
Application_ID	Application ID (PK)	INT
Candidate_ID	Candidate reference (FK)	INT
Job_ID	Job reference (FK)	INT
Applied_At	Submission timestamp	DATETIME
Status	Application progress	VARCHAR(30)

10 APPLICATION_STAGE

Field Name	Description	Data Type
Stage_ID	Stage ID (PK)	INT
Application_ID	Application reference (FK)	INT
Stage_Name	Name of recruitment step	VARCHAR(50)
Stage_Status	Stage result	VARCHAR(30)
Updated_At	Timestamp	DATETIME

11 INTERVIEW

Field Name	Description	Data Type
Interview_ID	Interview ID (PK)	INT
Application_ID	Application reference (FK)	INT
Interviewer_Name	Person conducting interview	VARCHAR(150)
Interview_Date	Date/time	DATETIME
Interview_Type	Type (online, in-person)	VARCHAR(50)
Result	Result	VARCHAR(30)

12 OFFER

Field Name	Description	Data Type
Offer_ID	Offer ID (PK)	INT
Application_ID	Application reference (FK, UQ)	INT
Offered_Salary	Offered amount	DECIMAL(10,2)
Start_Date	Proposed start date	DATE
Offer_Status	Status (sent/accepted/etc.)	VARCHAR(30)

13 RECRUITER_PERFORMANCE

Field Name	Description	Data Type
Performance_ID	Performance record ID (PK)	INT
Recruiter_ID	Recruiter reference (FK)	INT
Period_Month	Month of record	VARCHAR(20)
Total_Hires	Number of successful hires	INT
Avg_Time_To_Fill	Average time to fill positions	DECIMAL(5,2)

Summary

This recruiting agency database covers all major business processes — candidate management, job postings, applications, interviews, and offers — in a well-structured, **3NF-compliant** schema as mentioned criteria. The 3NF-compliant

schema is achieved as all fields are atomic, every attribute depends on full primary key and there are no transitive dependencies (attributes depend only on PK).

The ER-diagram is attached and represents logically connected, well-organized system which is ready to solve recruitment management challenges. The database is ready to be implemented into work.